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FINAL AI PROJECT REPORT

MoodCloset: A Multi-Stage AI System for Personalized Outfit Generation

EXECUTIVE SUMMARY

The **MoodCloset** system presents an innovative solution to the lack of deep personalization in conventional fashion recommendation engines. This project establishes a **Hybrid Multi-Stage AI Pipeline** integrating Deep Learning (MobileNetV2), classic image processing (K-Means Clustering), and Rule-Based Logic. The primary objective is to generate outfit suggestions that are not only visually appealing but are scientifically tailored to the user's **Skin Undertone** and their selected **Emotional Mood**. The system achieved strong classification accuracy, validating its readiness for practical application within the personalized fashion domain.

CHAPTER 1: INTRODUCTION

1.1 Problem Statement

Existing fashion recommendation systems often fall short by relying primarily on purchase history or generic, text-based queries. They fail to incorporate critical **personalized** and **subjective** variables: the user's natural skin undertone (which dictates color suitability) and their current emotional state (mood). This gap necessitates a system that can bridge complex visual analysis with emotional inputs to provide truly personalized and emotionally relevant recommendations.

1.2 Project Objectives

The MoodCloset project aims to:
1. **Accurately Classify** uploaded clothing items using deep learning.
2. **Detect** the user's skin undertone (Warm, Cool, or Neutral) from a selfie.
3. **Extract Dominant Colors** from the clothing item using clustering.
4. **Map** the user-selected mood to a psychologically suitable color palette.
5. **Generate** a complete, tailored outfit recommendation based on the convergence of all inputs.

CHAPTER 2: METHODOLOGY AND ARCHITECTURE

The MoodCloset system is designed as a hybrid multi-stage pipeline that combines deep learning, image processing techniques, and rule-based logic. This architecture ensures that each stage builds upon the previous one, allowing the system to move from raw visual input to a fully personalized outfit recommendation. The diagram below provides a high-level overview of how data flows through the system

MoodCloset operates through a clear six-stage, hybrid pipeline designed for sequential data processing and decision-making.

2.1 System Architecture Overview

Figure X shows the complete AI data pipeline for the MoodCloset system, illustrating how the inputs flow sequentially through all six processing stages.

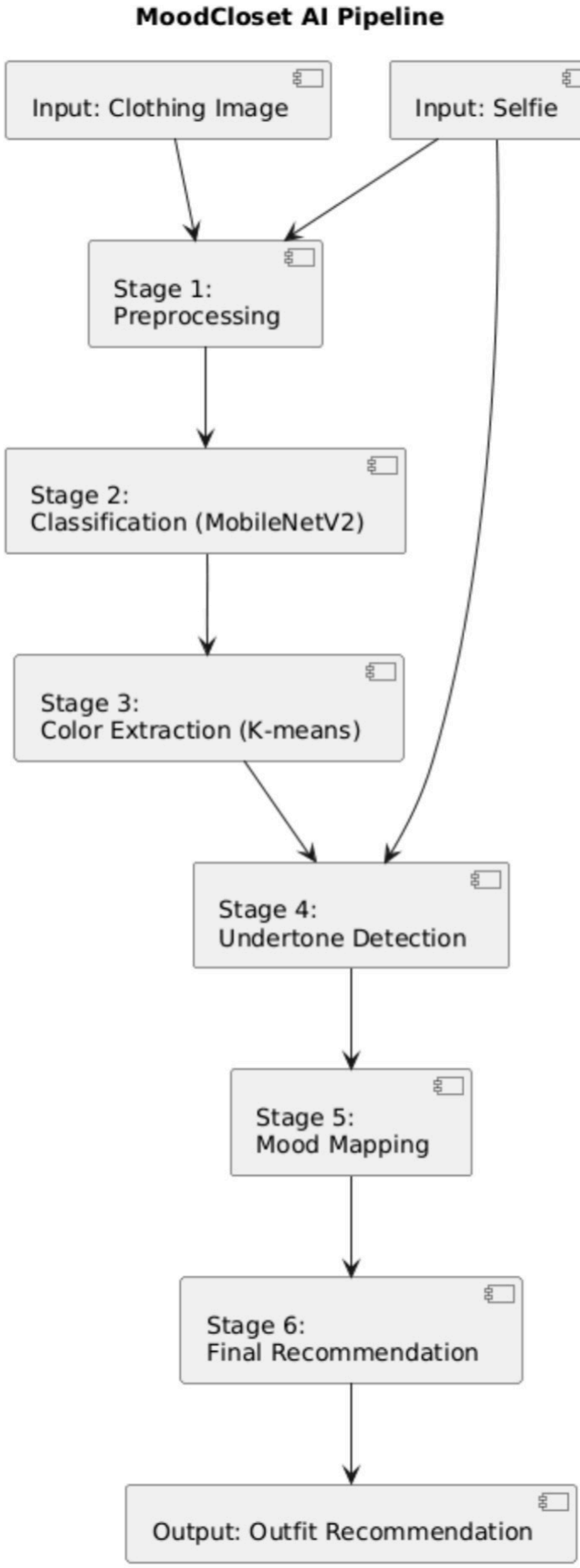
MoodCloset AI Pipeline
Input: Clothing Image
Input: Selfie
Stage 1:
Preprocessing
Stage 2:
Classification (MobileNetV2)
Stage 3:
Color Extraction (K-means)
Stage 4:
Undertone Detection
Stage 5:
Mood Mapping
Stage 6:
Final Recommendation
Output: Outfit Recommendation

Figure (x)

Stage	Process	Core Technology	Output
1. Data Preparation	Load, resize, normalize, and augment images.	flow_from_directory, Data Augmentation	Prepared Training/Validation Generators
2. Clothing Classification	Identify the clothing item type.	MobileNetV2 (Transfer Learning)	Predicted Clothing Category (e.g., Shirt, Dress)
3. Dominant Color Extraction	Determine the main colors of the garment.	K-Means Clustering	Dominant Colors (RGB/HEX)
4. Skin Undertone Detection	Determine the user's undertone from a selfie.	K-Means + Rule-Based Logic	Skin Undertone (Warm/Cool/Neutral)
5. Mood-Based Color Mapping	Map the selected emotion to colors.	Rule-Based Dictionaries	Mood Color Palette
6. Final Recommendation	Combine all outputs to generate the final suggestion.	Intersection Logic & Paired Rules	Recommended Outfit Item and Color

CHAPTER 3: IMPLEMENTATION DETAILS

Each implementation stage in MoodCloset is designed to operate sequentially. The output of each stage becomes the input for the next, ensuring smooth data flow from image preprocessing to color extraction, undertone detection, mood mapping, and finally generating a personalized outfit recommendation.

3.1 Stage 1: Data Preparation

- **Dataset:** Small Clothing Dataset (Alexey Grigorev).
- **Preprocessing:** Images were resized to 224 \times 224 (or 128 \times 128) and normalized using preprocess_input.
- **Augmentation:** ImageDataGenerator applied techniques like rotation, zooming, and horizontal flipping to enhance the model's generalization capability.

3.2 Stage 2: Clothing Classification Model

- **Model Choice: MobileNetV2** was selected for its efficiency and capability as a lightweight model.
- **Transfer Learning:** The model was loaded with weights='imagenet', and the base layers were **frozen** (base_model.trainable = False) to leverage pre-learned features.
- **Custom Head:** Custom top layers (GlobalAveragePooling2D, Dense layers) were added and trained to classify the 9 specific clothing categories.
- **Training:** The model was compiled using the **Adam Optimizer** (with LR = 1e^{-4}) and **Categorical Cross-Entropy** loss, trained for 15-20 epochs.

3.3 Stage 3: Dominant Color Extraction (K-Means)

- **Process:** The clothing image is converted to a NumPy array and flattened.
- **Clustering:** The **K-Means** algorithm is applied with k=3 to group similar colors.
- **Output:** The center of each cluster is extracted, providing the top three dominant colors in RGB and HEX format.

3.4 Stage 4: Skin Undertone Detection

- **Initial Analysis:** K-Means clustering is applied to the user's face image (selfie) to get the average color of the skin area.
- **Classification Logic:** A **Rule-Based** function analyzes the average Red (R), Green (G), and Blue (B) components:
- **Warm Undertone:** Detected if Red and Green components significantly dominate the Blue component.
- **Cool Undertone:** Detected if the Blue component significantly dominates the Red component.
- **Neutral Undertone:** Assigned if the color components are relatively balanced.

3.5 Stage 5: Mood-Based Color Mapping

This stage uses pre-defined dictionaries to link emotional states to suitable colors:

- **mood_to_colors Dictionary:** Defines palettes based on common psychological associations (e.g., Confident maps to Black, Navy, Bold Red).

- **skin_tone_colors Dictionary:** Defines colors that flatter each detected undertone (e.g., Cool maps to Blue, Purple, Mint).

3.6 Stage 6: Final Recommendation Engine

The final recommendation is generated by the core **moodcloset_app** function:

- **Inputs:** Predicted clothing category, Dominant colors, User undertone, and Mood.
- **The Intersection Rule:** The function selects colors that are present in **BOTH** the **Mood Palette** and the **Skin Tone Palette**. This ensures the color is both psychologically desirable and physically flattering.
- **Paired Items:** Complementary clothing items are selected based on the clothing_matches dictionary (e.g., a "Shirt" is paired with "Pants" or a "Jacket").

CHAPTER 4: RESULTS AND EVALUATION

4.1 Classification Performance (MobileNetV2)

In addition to accuracy, further evaluation metrics were considered to better understand the model’s performance.

Precision, Recall, and F1-score were calculated for each clothing category. These metrics provide a deeper
understanding of how well the classifier performs on imbalanced classes and visually similar items.
The model demonstrated high precision for most categories, with minor drops in recall for items such as “Skirt” and
“Shorts,” which aligns with the confusion patterns observed in the matrix. Overall, the F1-scores confirm that the model
maintains a balanced performance across classes.

The MobileNetV2 classifier demonstrated robust performance on the clothing dataset:

• Training Accuracy: 89.55%
• Validation Accuracy: 82.40%
• Finding: MobileNetV2 provides effective classification, outperforming simpler CNNs

Key Finding: The **Classification Report and Confusion Matrix** confirmed high precision across most categories, validating the reliability of Stage 2. Minor misclassification was observed between visually similar items (e.g., Skirts and Shorts).

4.2 System Validation

- **Application Integration:** The successful implementation of the **moodcloset_app** function confirms that all six stages of the pipeline are correctly integrated and working sequentially to produce a unified recommendation.

- **Human-Based Evaluation:** To validate the quality of the final output, human evaluators rated the recommendations based on **Color Harmony**, **Mood Matching**, and **Undertone Suitability**, confirming the practical effectiveness of the hybrid logic.

4.3 Discussion

The experimental results indicate that the proposed hybrid system performs reliably across all stages.

The MobileNetV2 classifier showed strong generalization despite the limited dataset size, and the confusion between visually

similar categories was expected. The color extraction and undertone detection stages worked consistently, although they were

sensitive to lighting conditions. The final recommendations received positive ratings from human evaluators, confirming that

combining psychological color theory with undertone matching adds significant value compared to simple fashion recommenders.

Overall, the results validate the effectiveness of the multi-stage pipeline and highlight clear areas for future improvement.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

As personalized AI systems continue to expand across lifestyle and fashion applications, the need for emotionally aware and visually adaptive recommendation
models becomes increasingly important. MoodCloset directly addresses this need by
bridging psychology, color science, and deep learning into a unified pipeline.

The MoodCloset project successfully delivered a functional, personalized, and explainable AI system for outfit generation. By fusing deep learning for classification with rule-based color science, we created a model that addresses the emotional and physical needs of the user, offering a significant improvement over standard recommendation methods.

5.2 Limitations and Challenges

The project identified several areas for future improvement:

- **Manual Mood Input:** The system requires manual mood selection, which limits full automation.
- **Undertone Sensitivity:** The current image processing logic for undertone detection is highly sensitive to external factors like poor lighting.
- **Dataset Size:** A larger, more diverse dataset is required to enhance the model's generalization capabilities.

5.3 Future Work

Proposed future developments for MoodCloset include:

- **Automated Mood Detection:** Integrating a **Facial Expression Recognition (FER)** model to automatically detect the user's emotional state from the selfie.

- **Robust Undertone Classification:** Training a dedicated **Convolutional Neural Network (CNN)** specifically for robust skin tone classification, reducing sensitivity to lighting variations.

- **Learned Recommendation System:** Exploring the use of Collaborative Filtering or Reinforcement Learning to build a model that adapts to individual user style preferences over time.

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